Professional Practice Standards in Alberta Education

Biostatistical consulting project

Integrated statistical evidence on implementation, professional learning need, contextual differences, structural alignment, and longitudinal trends

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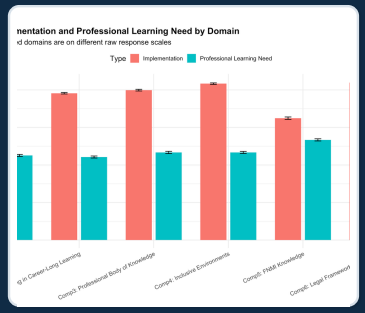
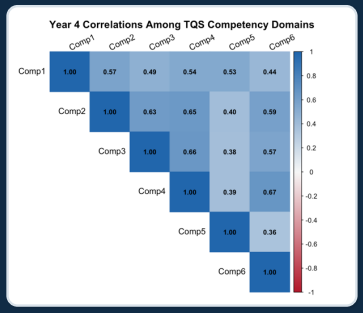
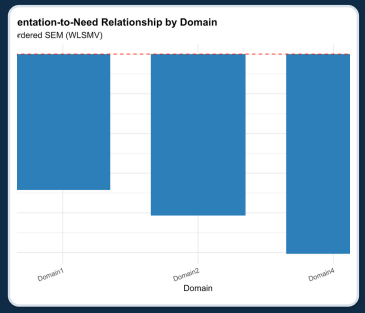
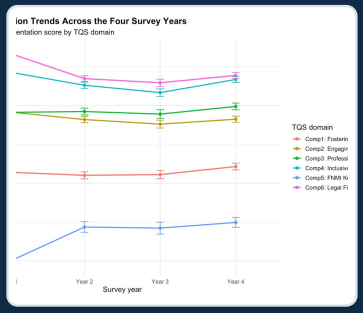








Project Outline:

- Descriptive profile
- Group comparison
- SEM
- Longitudinal positioning

Executive message

What the integrated evidence says at a glance

Bottom line

Alberta's implementation landscape is broadly strong and highly stable. The evidence points to one persistent priority area and supports targeted, not uniform, provincial action.

Integrated project signal

The strongest policy message is continuity with a focused improvement need in FNMI Knowledge.

1 Stable profile

Across four survey years, the relative ordering of domains changes very little.

2 Persistent priority

FNMI Knowledge remains the lowest-implementation and highest-need domain in Year 4.

3 Small contextual effects

Group differences are statistically detectable but modest in magnitude.

4 Aligned mechanism

Stronger reported implementation is linked to lower learning need in estimable domains.

Why this matters to Alberta

The analysis is designed to support provincial interpretation, prioritization, and follow-through

Professional Practice Standards are a policy anchor for teaching quality, professional growth, and equitable learning environments. Government decision-makers need evidence on where implementation is already strong, where support need remains elevated, and whether Year 4 reflects a durable system pattern or a one-year fluctuation.

- Describe current implementation across the six Teacher Quality Standard domains
- Identify where professional learning demand is concentrated
- Test whether differences across professional contexts are practically meaningful
- Show whether the Year 4 pattern is structurally coherent and longitudinally durable

How strong is implementation?

System performance

Which domains are consistently strong, and which remain weaker?

Where is the need signal highest?

Support targeting

How closely does learning need align with implementation gaps?

Can government act on Year 4?

Policy continuity

Does Year 4 reflect an enduring project trajectory rather than an anomaly?

Integrated analytical design

Four complementary workstreams were used to build a single policy-facing evidence base

A Descriptive profile

Domain means across Years 1–4, Year 4 implementation vs need, inter-domain correlations, and descriptive reliability.

B Contextual comparison

MANOVA/ANOVA across grade level, specialization, teaching experience, and survey year.

Integrated policy-facing evidence base

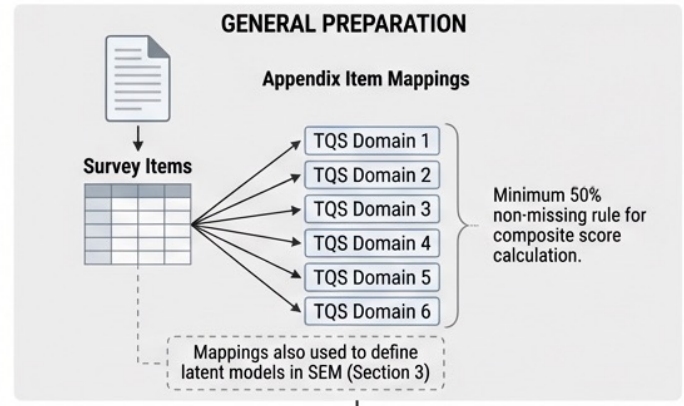
C Structural alignment

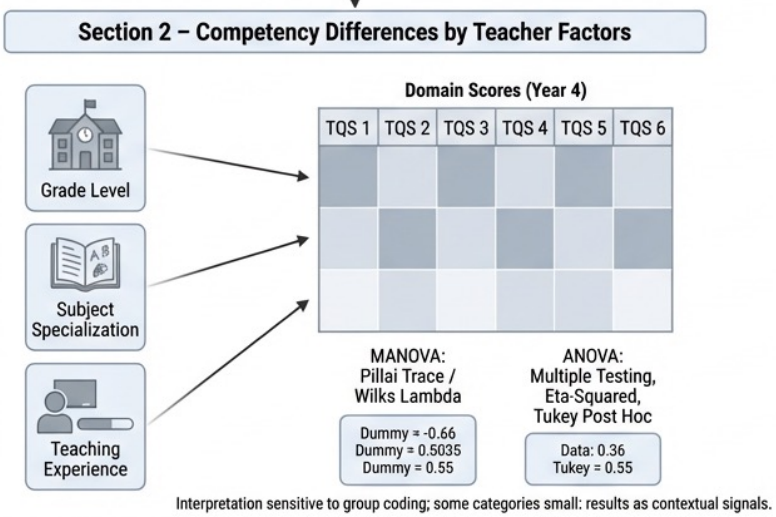
Domain-specific SEMs testing whether stronger implementation aligns with lower professional learning need.

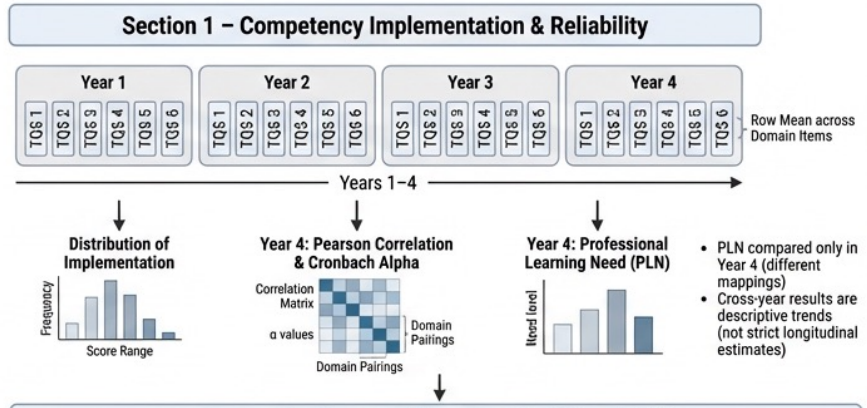
D Longitudinal positioning

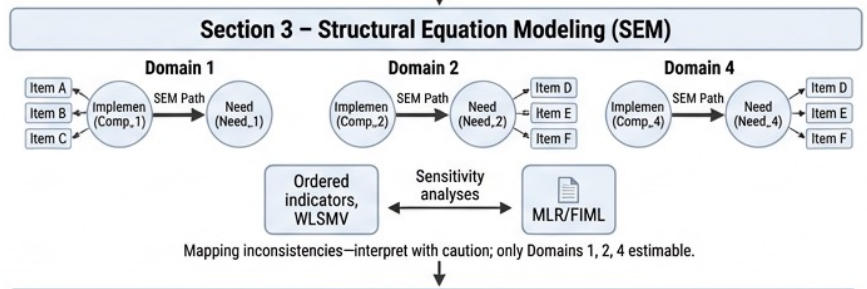
Year effects, rank stability, and profile correlations placing Year 4 in a four-year trajectory.

Methods


GENERAL PREPARATION
Appendix Item Mappings
Survey Items
TQS Domain 1
TQS Domain 2
TQS Domain 3
TQS Domain 4
TQS Domain 5
TQS Domain 6
Minimum 50% non-missing rule for composite score calculation.
Mappings also used to define latent models in SEM (Section 3)


Section 2 – Competency Differences by Teacher Factors
Grade Level
Subject Specialization
Teaching Experience
Domain Scores (Year 4)
TQS 1
TQS 2
TQS 3
TQS 4
TQS 5
TQS 6
MANOVA: Pillai Trace / Wilks Lambda
Dummy = -0.66
Dummy = 0.5035
Dummy = 0.55
ANOVA: Multiple Testing, Eta-Squared, Tukey Post Hoc
Data: 0.36
Tukey = 0.55
Interpretation sensitive to group coding; some categories small: results as contextual signals.

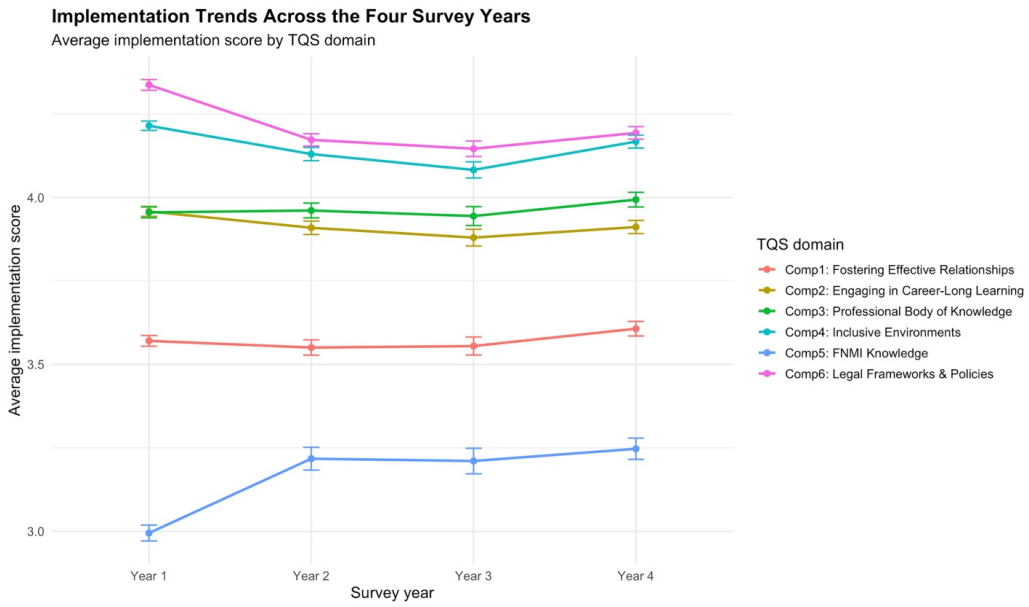

Section 1 – Competency Implementation & Reliability
Year 1
Year 2
Year 3
Year 4
Row Mean across Domain Items
Years 1–4
Distribution of Implementation
Score Range
Year 4: Pearson Correlation & Cronbach Alpha
Correlation Matrix
α values
Domain Pairings
Year 4: Professional Learning Need (PLN)
PLN compared only in Year 4 (different mappings)
Cross-year results are descriptive trends (not strict longitudinal estimates)


Section 3 – Structural Equation Modeling (SEM)
Domain 1
Domain 2
Domain 4
Item A
Item B
Item C
Implemen (Comp_1)
SEM Path
Need (Need_1)
Implemen (Comp_2)
Need (Need_2)
Item D
Item E
Item F
Implemen (Comp_4)
Need (Need_4)
Ordered indicators, WLSMV
Sensitivity analyses
MLR/FIML
Mapping inconsistencies—interpret with caution; only Domains 1, 2, 4 estimable.

Section 4 – Trends in Professional Learning Needs
Annual Implementation Scores
PL Need
Year 1
Year 2
Year 3
Year 4
Compare Year 4 with previous trends (Rank Order, Correlation)
PLN not longitudinal: descriptive summary only.
Implementation: repeated cross-sectional, not within-person change.
Year 1–3 implementation supported; multi-year PLN mapping not available.

Across four survey years, the implementation profile is highly stable

Legal Frameworks & Policies and Inclusive Environments remain strongest; FNMI Knowledge remains lowest



Appendix Figure D1

4.19 / 4.17

Highest Year 4 implementation

Legal Frameworks & Policies and Inclusive Environments remain at the top of the profile.

3.00 → 3.25

Persistent challenge

FNMI Knowledge improved from Year 1 to Year 4, but it remained the lowest domain in every year.

Comp4, Comp5, Comp6

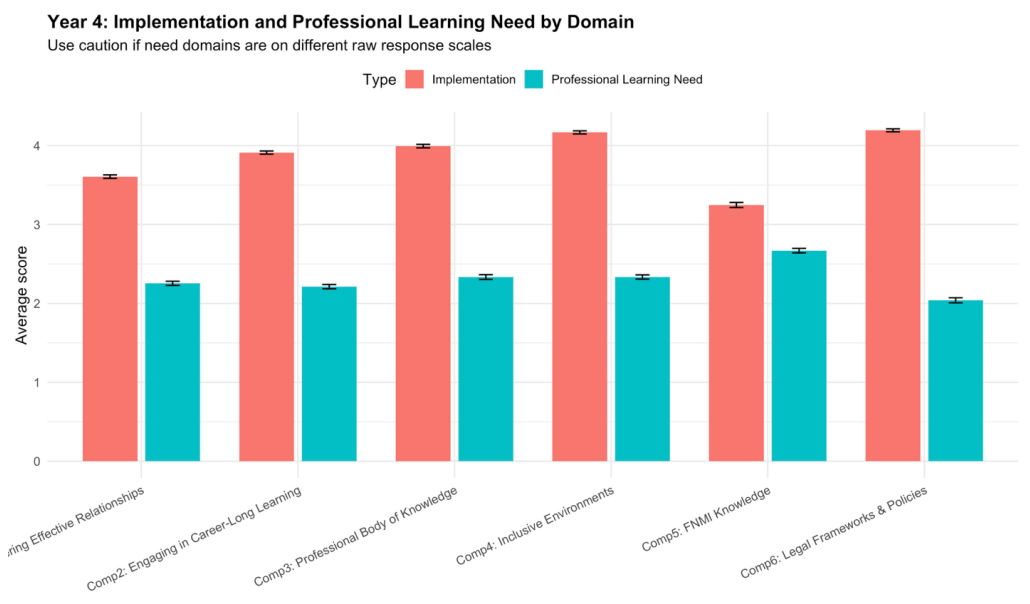
Where year effects appear

Formal year-related differences are concentrated in Inclusive Environments, FNMI Knowledge, and Legal Frameworks & Policies.

Interpretation: Year 4 does not represent a new implementation landscape. It is best interpreted as a later point on a broadly stable trajectory.

The clearest priority signal is FNMI Knowledge

The weakest implementation domain is also the highest-need domain in Year 4



Appendix Figure A2

2.67

Highest Year 4 need

FNMI Knowledge carries the strongest professional learning signal in the portfolio.

2.04

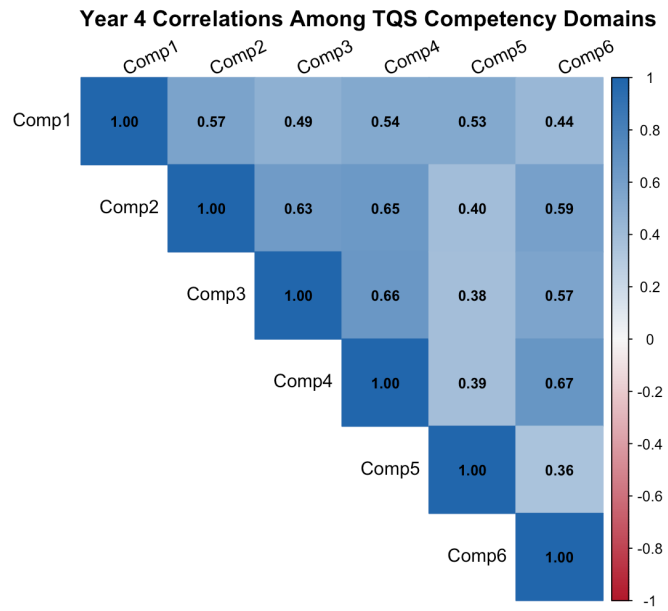
Lowest Year 4 need

Legal Frameworks & Policies combines strong implementation with the lowest reported need.

Policy implication
Targeted support for FNMI Knowledge is better justified than a uniform intervention across all domains.

The six domains behave as a connected professional practice profile

Relationships are positive across the portfolio, but FNMI Knowledge is less integrated than the rest



Appendix Figure A3

r = 0.668

Strongest association

Inclusive Environments and Legal Frameworks & Policies form the tightest pair in the Year 4 profile.

0.411

Least interconnected domain

FNMI Knowledge has the lowest average absolute correlation with the rest of the portfolio.

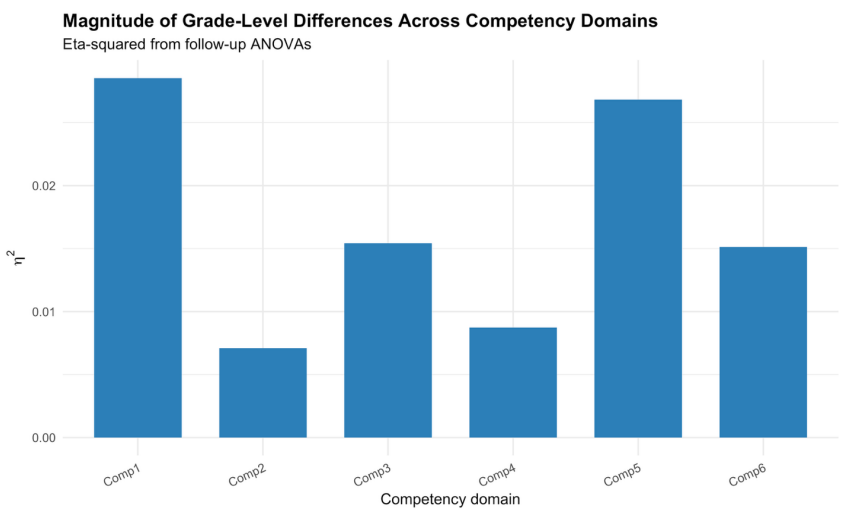
α = 0.68 to 0.95

Reliability range

The Year 4 composites are internally coherent enough to support integrated interpretation.

Contextual differences exist, but effect sizes remain small

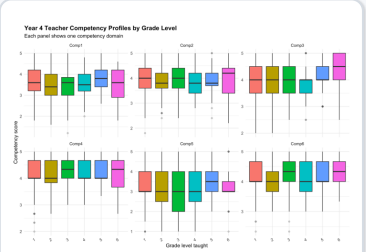
FNMI Knowledge is the most context-sensitive domain; teaching experience shows the clearest practical signal



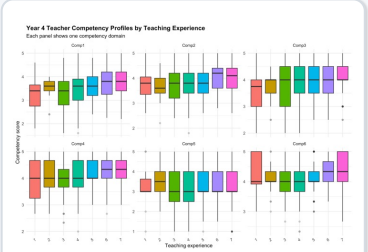
Appendix Figure B4

Key interpretation

- All detected contextual effects are small in magnitude (η^2 values remain low across domains).
- Teaching experience produces the strongest multivariate signal (Pillai trace = 0.062), but it is still modest.
- FNMI Knowledge shows the most repeated sensitivity across grade level, specialization, and year.



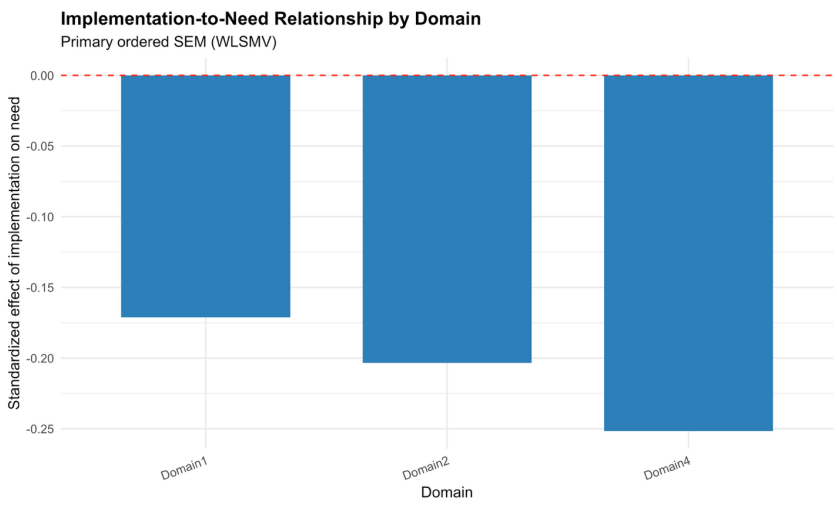
Appendix Figure B1



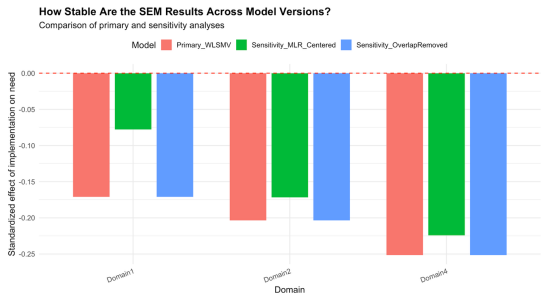
Appendix Figure B2

Stronger implementation is consistently associated with lower professional learning need

The inverse relationship is strongest in Inclusive Environments and remains stable across sensitivity checks



Appendix Figure C5



Appendix Figure C6

-0.25
Domain 4
Inclusive Environments: strongest and most robust inverse relationship

-0.20
Domain 2
Engaging in Career-Long Learning: strong and stable inverse relationship

-0.17
Domain 1
Fostering Effective Relationships: interpretable but more sensitive

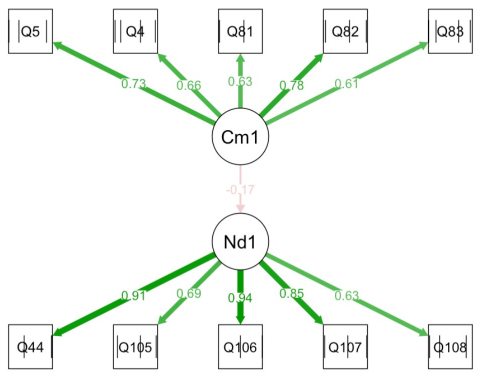
Boundary: only Domains 1, 2, and 4 were estimable under the current mapping and data structure.

Stronger implementation is consistently associated with lower professional learning need

The inverse relationship is strongest in Inclusive Environments and remains stable across sensitivity checks

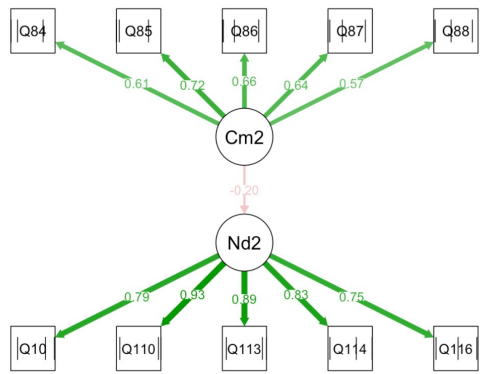
Supporting figures

Appendix Figure C1. Structural equation model for Fostering Effective Relationships, showing the measurement structure for implementation and need indicators in Domain 1.



Appendix Figure C1

Appendix Figure C2. Structural equation model for Engaging in Career-Long Learning, showing the measurement structure for implementation and need indicators in Domain 2.

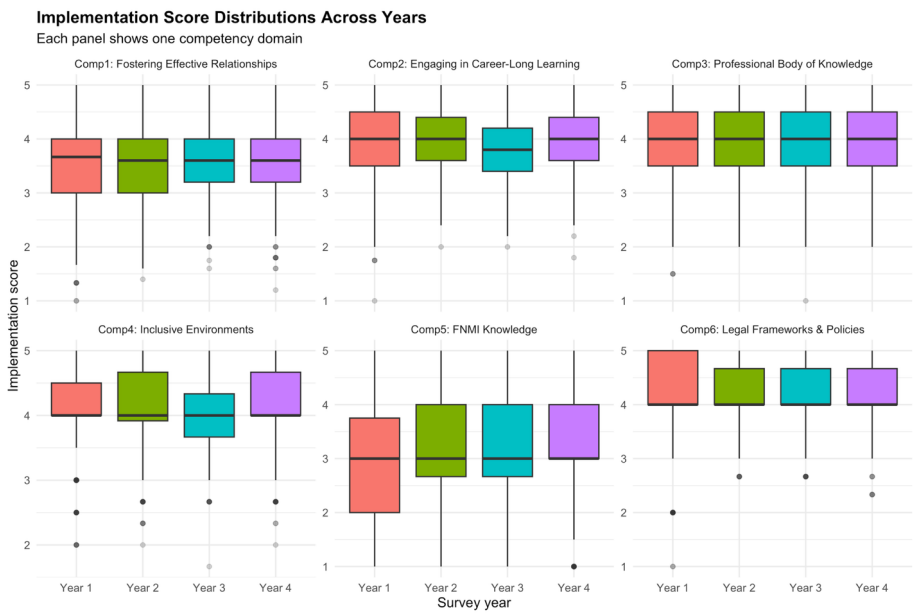


Appendix Figure C2

Boundary: only Domains 1, 2, and 4 were estimable under the current mapping and data structure.

Year 4 aligns closely with the broader project trajectory

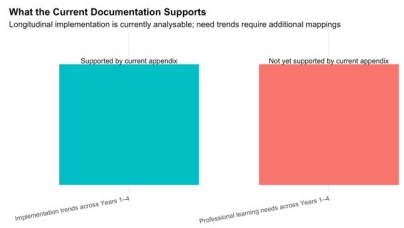
The same strength–challenge ordering persists across all four survey years



Appendix Figure D2

Profile consistency with Year 4

Year 4 vs Year 1	Year 4 vs Year 2	Year 4 vs Year 3
0.9936	**0.9989**	**0.9990**



Appendix Figure D4

Interpretation: The Year 4 pattern is part of an enduring repeated cross-sectional trajectory. Multi-year need trends remain unavailable because explicit need mappings exist only for Year 4.

Policy implications

The evidence favors targeted improvement inside a broadly strong implementation landscape

1 Protect system strengths

Sustain strong implementation in Legal Frameworks & Policies and Inclusive Environments through continued reinforcement rather than major redesign.

2 Prioritize FNMI Knowledge

Treat FNMI Knowledge as the clearest targeted improvement priority because it is persistently lowest in implementation and highest in learning need.

3 Differentiate supports

Use contextual signals for targeted support design. The evidence does not justify a uniform province-wide intervention across all teacher groups.

4 Strengthen technical monitoring

Preserve the integrated workflow and improve longitudinal documentation for professional learning need so future monitoring can move from repeated cross-sectional evidence to fuller trend tracking.

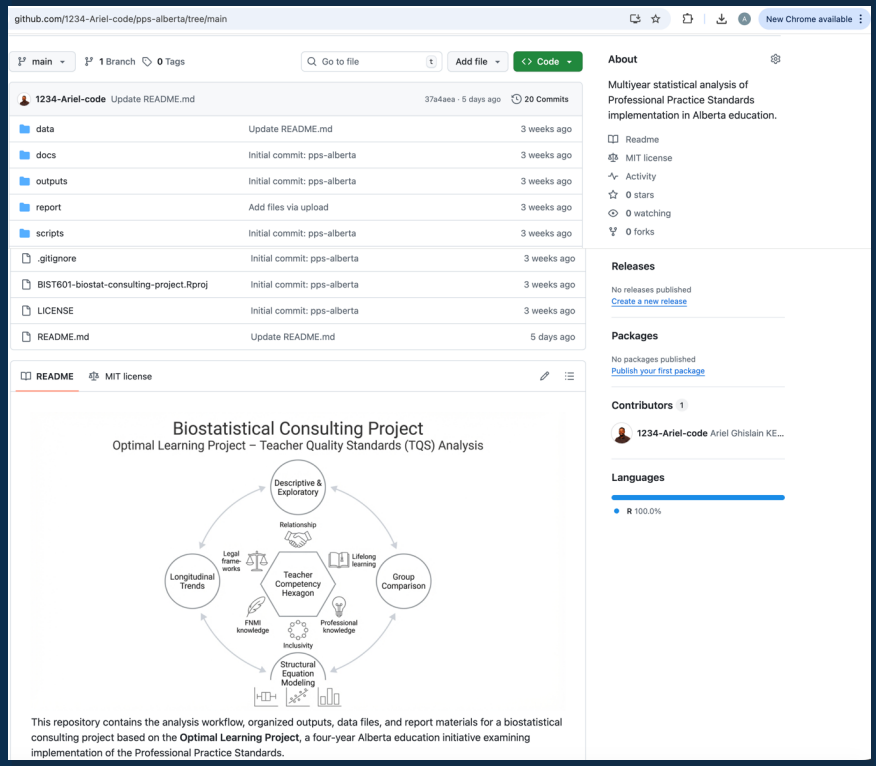
Overall conclusion: the province is seeing broad implementation strength, a persistent FNMI Knowledge gap, coherent alignment between stronger practice and lower support need, and small contextual differences best addressed through targeted action.

Thank you

Technical access and reproducibility

Public repository
github.com/1234-Ariel-code/pps-alberta

- Includes analysis scripts, project structure, and reproducibility materials
- Supports technical review of the workflow used to generate the report outputs
- Underlying data remain subject to permissions, privacy requirements, and confidentiality restrictions



Questions and discussion